

Authenticity and academic integrity in Generative Artificial Intelligence (GenAI) use among undergraduate nursing students

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ABSTRACT

Background: The use of large Generative Artificial Intelligence (GenAI) in higher education is reshaping how students learn. While GenAI offers significant benefits and the potential to improve education, it also poses ethical challenges and may affect students' ability to identify learning gaps and engage with course content.

Purpose: This study aims to explore students' experiences with GenAI related to academic integrity and the authenticity of work, including their awareness and attitudes toward acceptable use.

Method: This study employed a qualitative descriptive approach. The participants were recruited from the Bachelor of Science (Nursing) program in an academic university during the period October 2024 to January 2025.

Results: Fourteen undergraduates in their second- and third-year of study participated in group interviews. The qualitative data analysis revealed five key themes: 1) integration with learning, 2) trust and credibility of GenAI information, 3) ethical considerations, 4) addressing academic integrity, and 5) guidance on using GenAI.

Conclusion: These five key themes underscore the need to better understand the role of GenAI in higher education, particularly in self-directed learning and managing learning activities. With regards to academic integrity, clear guidelines must be put in place for educators to guide the use of GenAI, emphasizing its use as an instrument to augment, and not replace original work.

Introduction

The integration of Generative Artificial Intelligence (GenAI), particularly Large Language Models (LLMs), is significantly transforming education across various fields, including healthcare disciplines like nursing (Janumpally et al., 2025). Established as a discipline in 1986, artificial intelligence focuses on creating systems capable of learning, reasoning, and performing tasks that typically require human intelligence (MeSH, 2024), while GenAI leverages advanced algorithms and data to emulate human cognitive processes, reshaping how nursing students interact with learning materials. LLMs are a part of GenAI and are defined by MeSH (2025) as computational designs that apply deep learning methods to extremely large datasets to predict and generate natural-sounding text, enabling the responsive, information-rich learning experiences described above.

Nursing students are increasingly incorporating GenAI tools to enhance their learning experiences, traditionally reliant on clinical exposure and evidence-based knowledge. GenAI features boost student engagement and motivation by offering real-time responses and personalized feedback, creating dynamic, tailored learning environments (Li et al., 2024). In practice, students use GenAI to interact with virtual patients for assessment and clinical reasoning, rehearse procedure skills with step-by-step checklists and practice communication handovers with immediate feedback on clarity, completeness, and safety. This encourages deeper exploration of course materials, fostering active learning and transforming students into active participants in their educational journey (Glauberman et al., 2023). It also stimulates curiosity by providing instant feedback such as real-time explanations of concepts, reinforcing learning and motivation (Abd-alrazaq et al., 2023). The ability to receive fast, personalized responses enhances

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student interest, even in complex or challenging topics like pathophysiology and ethical decision-making (Janumpally et al., 2025), empowering them to take control of their learning and potentially improving educational outcomes (Li et al., 2024). Additionally, GenAI assists students by suggesting evidence-based resources and highlighting gaps through post-activity debrief analyses, effectively addressing knowledge deficiencies (Abd-alrazaq et al., 2023).

Despite the numerous advantages offered by GenAI, there is a growing concern regarding nursing students' over-reliance on these GenAI tools. This dependency stems from the convenience of instant answers and explanations provided by GenAI, which can lead to reduced engagement with traditional literature and resources. Consequently, students may miss opportunities to conduct in-depth reviews and analyses, which are crucial for fostering self-reflection and critical evaluation which are key components in recognizing personal learning needs. The ease of accessing quick solutions through GenAI can diminish the necessity for students to engage in reflective thinking or thoroughly assess their understanding (Ahmad et al., 2023). This reliance might impede the development of self-regulation skills that are essential for continuous professional growth in the nursing field. Without engaging in deeper cognitive processes to identify learning gaps, students risk overlooking opportunities for meaningful self-assessment and problem-solving (Glaberman et al., 2023).

While GenAI offers substantial benefits and has the potential to enhance educational experiences, it also presents ethical challenges. These challenges are particularly related to academic integrity, the authenticity of student work, and the potential impact on acquiring fundamental nursing competencies (Currie, 2023). Furthermore, the interactive nature of GenAI technologies raises concerns about their influence on students' ability to effectively identify learning gaps and engage with course content (Glaberman et al., 2023). Given the novelty of GenAI, there is limited understanding of its comprehensive impact on nursing students' learning experiences. Therefore, this study aims to explore students' experiences with GenAI related to academic integrity and the authenticity of work, including their awareness and attitudes toward acceptable use.

Method

Design

This qualitative study, conducted as part of a larger quantitative investigation, was conducted to provide a direct description of the students' experience with GenAI using purposeful sampling and thematic analysis to generate rich, descriptive insights and not intended to correlate with or directly compare to the quantitative results. The Consolidated criteria for reporting qualitative research (COREQ) guidelines were used for transparent reporting of the nursing students' learning experiences (Tong et al., 2007).

Setting and participants

Ethics approval was obtained from the National University of Singapore ethics committee (IRB-2024-558). The application encompassed both quantitative and qualitative components collected and analyzed independently to achieve the aims. Participants were Bachelor of Science (Nursing) undergraduates in their second- or third-year of a three-year nursing program with an additional Honors year contingent on meeting a specified GPA threshold, selected for the study to ensure sufficient exposure to nursing coursework and assessment practices relevant to the study's aim. First-year students were excluded because their limited curricular experience would likely constrain their ability to reflect meaningfully on GenAI's role in learning and academic integrity. Eligibility also required any prior use of a LLM, with no minimum frequency or duration specified, and was inclusive of all LLM platforms, which are services that let people and software interact with powerful

artificial intelligence text models, which are services that let people and software interact with powerful artificial intelligence text models to ask questions and generate text, to ensure that they could speak from direct experience with LLMs while avoiding bias toward any single product or ecosystem. Data was collected between October 2024 and January 2025. Focus groups were conducted until data saturation was achieved (Guest et al., 2006).

Procedures

In the first week of the academic year, undergraduate students were invited to participate in the study via email sent by the university administrative staff to minimize undue pressure on students to participate. Interested participants to be interviewed were asked to provide informed consent and to record their demographic characteristics via the online link (Qualtrics, 2025). The study invitation emphasized voluntary participation with no impact on grades or faculty relationships and no further follow-up. Recruitment occurred outside class hours, with no faculty attendance records and informed consent via were provided an online link containing independent contacts to ethics committee for confidential queries or concerns about coercion or retaliation and students could withdraw anytime.

Participants' names, contact information, and emails were collected, and focus group interview were conducted using semi-structured questions through online platforms (Zoom) allowing students to articulate their experiences candidly. The focus groups were facilitated by the principal investigator, who was not involved in the students' assessment or grading.

To protect the confidentiality and integrity of the interviews, security measures such as encryption and meeting access controls were implemented to prevent data interception. Participants were assigned pseudonyms (e.g., P1, P2) for anonymity in reports and presentations where interview quotes were used. Students who agreed to interviews provided their contact details solely for arranging the sessions, and this information was discarded after transcript verification by participants.

Audio recordings of the interviews were made to ensure data accuracy and completeness, allowing the principal investigator to review the recordings for themes, patterns, and insights. Transcriptions were created from the recordings for detailed analysis, and the recordings were deleted once the transcripts were verified. Participants who declined recording were excluded from the study. The interview questions (Fig. 1) were developed to address the study aim. The interview questions were first derived deductively from the study aim and anchored in established literature on learning with educational technologies, drawing on domains commonly examined in higher education and nursing education (Cotton et al., 2023; Sallam, 2023; Zhai, 2023). We drafted open-ended, non-leading items to elicit concrete examples and reflections across these domains and prepared neutral probes to deepen responses. The guide was then pilot tested after three participants for an initial, iterative pilot of semi-structured interview questions to detect major comprehension and wording issues. Further review was made after the fifth participant to evaluate the need for substantial revisions (Hennink & Kaiser, 2022). The final guide retained the core domains in Fig. 1 and notes for iterative refinement during data collection. All interview questions were optional, and participants were not required to answer questions they were uncomfortable with.

Data analysis

Statistical analysis was performed using IBM SPSS Statistics 26 (IBM Corp., 2019). Descriptive statistics were applied to profile the participants, with categorical data presented through frequency distributions and continuous data, such as age, reported using means and standard deviations. All interviews were transcribed, and participants were invited to review these to ensure accurate interpretation of meanings. Content analysis of qualitative data was then conducted using both

Interview Questions	
The main interview questions are structured according to the aims of this study and given below. Interview questions will be adapted as the interviews and data analyses proceed.	
Introduction:	
1.	How familiar were you with large language models (LLMs) like before their integration into your coursework?
2.	Can you describe your initial impressions of using these tools in your studies?
3.	Can you share specific instances where LLMs significantly helped you in your coursework?
Self-directed learning	
4.	How has LLMs supported your ability to learn independently?
5.	Can you describe how you use LLMs to manage and direct your own learning activities?
6.	How has LLMs impacted your access to learning resources and information?
Students' ability to identify learning needs	
7.	How effective is LLMs in helping you understand complex nursing concepts and medical terminology?
8.	Do you feel that using LLMs has improved your ability to retain and recall information? How so?
Engagement and motivation.	
9.	Has using LLMs made you more engaged with your learning materials? How?
10.	Are there any features of LLMs that particularly enhance your interest in your studies?
11.	Do you feel more motivated to learn and explore topics on your own with LLMs? Why or why not?
12.	Have you noticed any changes in your grades or the quality of your assignments since using ChatGPT?
Ethical concerns regarding academic integrity and the authenticity of work	
13.	Are there specific resources or types of information you find easier to obtain through LLMs?
14.	How do you ensure that you use LLMs ethically and maintain academic integrity in your work?
15.	Have there been any concerns about plagiarism or over-reliance on LLMs among your peers?

Fig. 1. Interview questions.

NVivo software and manual coding to thoroughly uncover new themes and patterns (QSR International Pty Ltd., 2018).

The process began with the research team to familiarize themselves with the content through repeated readings, gaining a deep understanding of the students' experiences with GenAI (Nowell et al., 2017). Significant statements regarding their interactions with GenAI were identified and coded to capture key aspects of these experiences. To maintain credibility, the author and one of the co-authors employed bracketing, and reflexive journaling to consciously set aside their pre-conceptions, engaged in peer debriefing across the team. During the interview, the key points were paraphrased by the interview to verify the meaning before moving on. Therefore, the transcripts were not returned to participants to prevent the risk of "over-correcting" their original statements that might affect the authenticity of the data. To enhance clarity and flow, long quotes containing filler words were removed to reduce distraction in the in-text quotations.

All transcripts were transcribed verbatim and the research team engaged in familiarisation by reading and re-reading the dataset while noting initial analytic observations. Using an inductive, reflexive approach, the first author then conducted systematic coding across the entire dataset in NVivo, attending to both semantic and latent meanings; analytic memos and an evolving codebook documented emerging insights and decisions. Two members coded the first 20 % of fixed units (speaker turns excluding administrative chatter) for each transcript independently and the intercoder reliability (ICR) was 96.9 %, suggesting strong percent agreement between the two. Any discrepancies were resolved through consensus coding to finalize the codebook before the remaining 80 % of the data was analyzed. Codes were collated into candidate themes and subthemes through iterative clustering and visual mapping. Candidate themes were reviewed at two levels with the first against the collated coded extracts and then against the full dataset to evaluate coherence, distinctiveness, and scope where themes were refined, merged, split, or discarded as necessary (Supplementary File 1). Each theme was then defined and named to articulate its central organizing concept and boundaries, and vivid, context-rich quotations were selected to support the analytic narrative addressing the research questions. To enhance trustworthiness, we maintained a transparent audit trail of coding and analytic decisions in NVivo (dependability) and

preserved raw transcripts, coded segments, and reflexive memos (confirmability). Credibility and analytic rigor were supported through ongoing reflexive practice and peer debriefing: a co-author independently engaged with the data and interpretations were discussed critically to deepen meaning. Transferability was supported by providing thick description of the study context, sampling strategy, participant characteristics, and richly contextualized quotations, enabling readers to judge applicability to similar higher education and nursing education settings. Throughout, reporting adhered to the COREQ guidelines to ensure comprehensive and transparent presentation of methods and findings.

Results

Interview group sizes were determined according to participants' stated availability, with the aim of maximizing attendance while minimizing scheduling burden. However, on the day of data collection, one participant notified the team of a last-minute inability to join her scheduled session and requested an alternative time. To accommodate this request and avoid excluding her perspectives, a separate make-up interview was arranged using the same interview guide and procedures. In total, four sessions were conducted: one individual interview and three group interviews comprising two, four, and seven participants, respectively, following the identical protocol (setting, facilitator, consent, audio recording, and question sequence). The average age of the participants at the time of the interviews was 21.85 (19–30) years, with five males and nine females. Among them, majority of the participants completed the GCE A-Level (Cambridge), an internationally recognized pre-university qualification used for university admissions (Table 1). In total, four interview sessions were conducted, each lasting approximately 30 to 60 min, depending on the number of participants in each group.

The qualitative data analysis revealed five key themes. The first theme centered on incorporating GenAI into their learning process and examining how these tools were integrated into educational settings. The second theme addressed the reliability and trustworthiness of information provided by GenAI, highlighting concerns about the accuracy of GenAI content. Ethical implications formed the third theme, focusing on the broader ethical considerations of using GenAI in education. The fourth theme involved ensuring academic integrity, discussing the challenges and strategies for maintaining originality and honesty in academic work when using GenAI. Finally, the last theme provided guidance on utilizing GenAI effectively, offering directions and support for students and educators to use these tools responsibly and beneficially (Fig. 2).

Theme 1: integration with learning

The use of GenAI in education brings several benefits and challenges. Participants shared that they appreciate the prompt responses and convenience GenAI offer, providing quick access to information without the wait associated with faculty feedback.

Table 1
Participants' characteristics.

Characteristics	Overall
N	14
Age (M)	21.85 (SD = 2.770)
Gender	
Male	5 (35.7 %)
Female	9 (64.3 %)
Education level	
Diploma	4 (28.6 %)
Bachelor	1 (7.1 %)
A' levels	9 (64.3 %)

(M = Mean, SD = Standard deviation.)

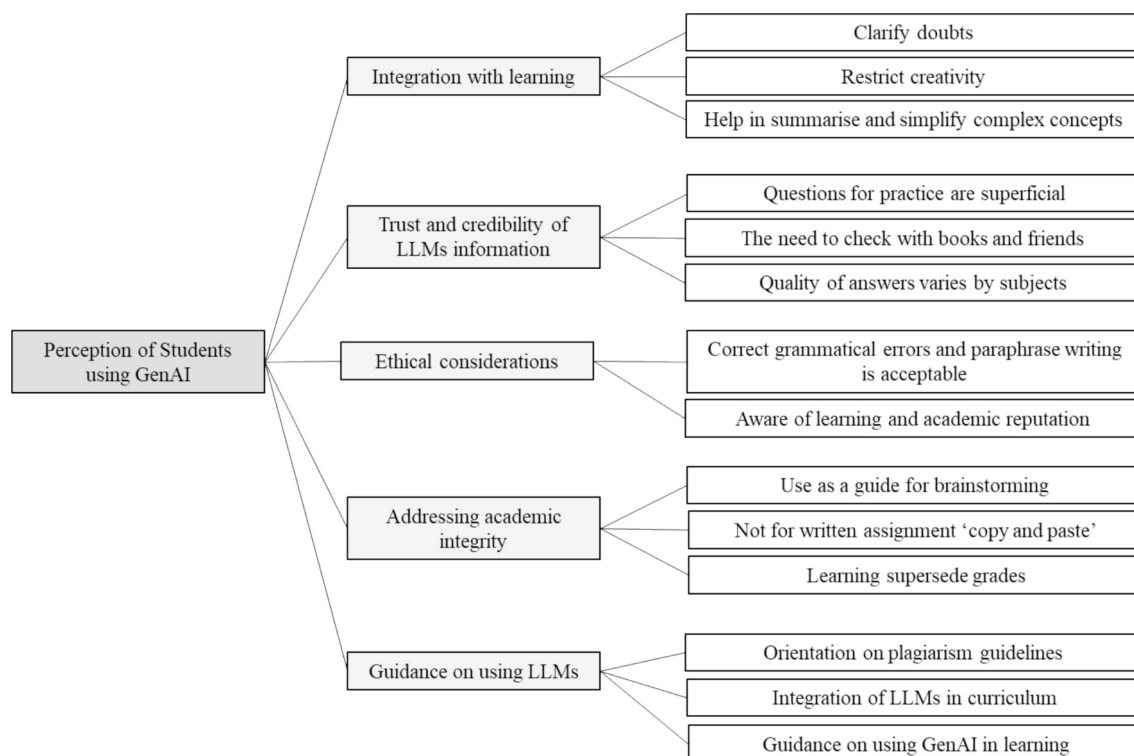


Fig. 2. Thematic map diagram. This figure illustrates the analytic structure of the findings, showing central themes, supporting subthemes, and their relationships developed during the analysis.

“I normally use ChatGPT to clarify my doubts and helps me to generate ideas because Google takes time to like sort out what you’re trying to find. But for ChatGPT, it’s like it gives you the answer straight away.”

– P4

Participants further described how GenAI enhanced their understanding and boosted their engagement and motivation by making complex topics more easily understood. They suggested that GenAI improves learning efficiency and productivity by streamlining concepts and ideas, aiding in tasks like writing assignments and exam preparation. Furthermore, GenAI assists in presenting complex concepts clearly, organizing information effectively, and analyzing data.

“One good thing about [ChatGPT] and also great, is summarizing data and organizing data in a way that’s easier for me to understand. So first of all, sometimes I’ll take article and put in the ChatGPT and ask them to list out the key points in this paper.”

– P2

However, participants shared how the immediacy of GenAI can impact creativity and critical thinking, raising concerns about students’ independent thought and information verification.

“I think for me, it restricts my creativity because let’s say I’m about to write my essay I already know that ChatGPT can be there to provide me with pointers about content and I tend to go straight to ChatGPT to see what it can offer me, and then I build from there. But more often than not, I’ll just take whatever they give me. So, like that definitely restricts me because I don’t bother using my own ideas.”

– P11

Theme 2: trust and credibility of GenAI information

Participants expressed concerns about the reliability of responses from GenAI, citing instances of incorrect or inaccurate information that

fostered skepticism toward the GenAI tool. Students recognized the need to cross-check ChatGPT’s answers with other sources, compare its information with personal notes, lectures, or seek additional references to ensure accuracy.

“Like, maybe from my notes, I’ll just like put into ChatGPT and ask them to ask me the questions back so, I can like at least get a base. I admit that some of the questions are quite simple so at least I can get a foundation of information that I need to know but some [ChatGPT] give wrong answers.”

– P3

While there is an expectation for GenAI to continue improving and impacting teaching and learning, the participants recognized the need to adapt to these technological changes. Notably, the quality of answers varies by subject, with some participants finding GenAI more reliable for summarizing some content but less so for others, warranting caution despite anticipated performance enhancements over time. Moreover, the participants highlighted the contrast between the accuracy of GenAI’s answers and the expected reliability of faculty-provided information or information found via traditional search engines.

“If I need more concise things, I’ll use ChatGPT but sometimes, if I need guiding questions when it’s nearing exam, to help generate some questions for me to practice, they might give wrong information. Sometimes I was testing it and I know the answer already but they give me wrong answer so I cannot like fully like 100% trust the answer that they give me. I still have to read through and look through the books, clarify with friends.”

– P6

Theme 3: ethical considerations

Participants discussed several aspects of ethical considerations when utilizing GenAI in academic contexts. They shared how they understood the accessibility of GenAI in providing a written work for their

assignment. However, they also emphasized the importance of taking responsibility for their actions and considering how unethical use could affect their learning and academic reputation. Participants agreed that using GenAI for purposes such as correcting grammatical errors and paraphrasing writing is acceptable when the primary ideas and thought processes originate from the students themselves.

“When you ask a question, let’s say for written assignment, most people [would] just put it into ChatGPT, and look at the answer given, and see where they can or how they can use the outline given to start their research. To me, I don’t have an issue with that. It’s about how you use information in the ethical way to get the information they want. So what’s not right is, you copy and paste the whole thing like into your assignment.”

– P1

Participants further discussed whether students should be penalized or graded for using GenAI, with educators and institutions crucial in guiding its use. Personal goals, like achieving high scores or gaining expertise, influenced how students used GenAI, necessitating a balance between leveraging GenAI for performance and maintaining academic integrity. Participants also shared how grades are not their main concern, valuing the completion of their own work over comparisons, and highlighting the importance of integrity and self-reliance.

“I don’t think it [grades] matters too much, because, in the first place, I don’t really like compare grades. At the end of the day for me, the fact that I do my own work. That’s more than enough for me.”

– P12

Theme 4: addressing academic integrity

Participants advocated for the integration of GenAI into regular coursework and assignments, suggesting that specific courses be created to incorporate and evaluate their use. However, they also emphasized the need for strategies that motivate students to explore and reflect on the use of GenAI, highlighting the role of self-reflection in understanding their benefits and limitations, viewing information from GenAI as a starting point for brainstorming rather than something to use verbatim, and cautioning against copying and pasting entire essays.

“I think it’s really self-regulating. You know that the information is there to help you and it’s not for you to take it as it is. So, it’s just either a starting point and brainstorm process. It’s not like you take it, and then you use it. For essays, some people copy and paste like the entire essays ... just don’t do that.”

– P10

When asked about concerns regarding peers obtaining higher scores with the help of GenAI, students emphasized their preference for valuing long-term learning processes and integrity. They remarked that those who rely on GenAI for assistance in writing and submitting work must answer to their own conscience. The students expressed that they cannot control such actions and choose instead to focus on their own sense of integrity.

“I think for me it’s more of like a personal conscience. If I use it to complete my work, I wouldn’t feel good about it. So, like as much as I feel like it’s unfair that they might get a higher score. That doesn’t really matter to me.”

– P8

Theme 5: guidance on using GenAI

Participants explained that the widespread availability and growing sophistication of technologies like LLM platforms posed challenges in ensuring their proper use by students. They seek the need for clear

guidelines and examples to clarify what constitutes plagiarism and the ethical use of GenAI.

“I think there [school] should be clear cut guidelines, because right now they’re just saying, Oh, you shouldn’t use AI ... It would be nice to have clear examples cause I sometimes also not really sure, like what would be considered as plagiarism and what is not plagiarism ... so it gets confusing.”

– P3

The participants appreciate the integration of GenAI into their curriculum to make their learning more engaging and interesting by presenting information in diverse ways and simplifying complex concepts to improve understanding. However, they felt left to explore topics on their own without guidance and expressed a need for tutors to help contextualize the knowledge into practice.

“Integration of AI and ChatGPT to the nursing education curriculum, I think, is okay. We can integrate the AI to our learning but you [tutors] cannot just expect students to just purely learn from AI and ChatGPT. I think that AI can assist but not help to completely teach. I think it needs to tie in with what you will actually face in our nursing workforce.”

– P9

Discussion

The objective of this study was to investigate undergraduate students’ perceptions and experiences of using GenAI in related to academic integrity and the authenticity of work, including their awareness and attitudes toward acceptable use. Participants in our sample perceived that GenAI tools influenced their ability to identify their learning needs, independently seek information, and manage their learning activities. They also described the interactive nature of GenAI as enhancing their engagement, motivation, classroom participation, assignment completion, and enthusiasm for learning. These perceptions align with prior research suggesting that GenAI can support self-directed learning by providing immediate, personalized guidance on complex medical concepts, clarifying misunderstandings, and tailoring explanations to learners’ needs (Li et al., 2024; Shalong et al., 2024). Participants in our study further indicated that GenAI facilitated exploration beyond prescribed course materials, which resonates with reported perspectives in the literature (Gunawan et al., 2024).

Despite these benefits, participants were concerned that over-reliance on GenAI-generated answers may impede their ability to engage in reflective thinking. Consistent with prior findings, they acknowledged the need for personal accountability when using GenAI and expressed awareness of potential repercussions of unethical practices for their learning and academic reputation (Choudhury & Chaudhry, 2024; Lin & Chen, 2024). Some participants also cautioned tensions between performance goals (e.g., high grades) and deeper learning, noting that insufficiently balanced use of GenAI could lead to submitting AI-produced work without adequate personal engagement, thereby risking plagiarism and undermining academic honesty (Gruenhagen et al., 2024). From participants’ viewpoints, these concerns underscore broader questions of responsibility and ethical conduct in technology-supported learning, where originality and evidence-based practice remain central (Pressman et al., 2024). Aligned with this perspective, previous studies emphasize clarifying ethical versus unethical uses of GenAI, encouraging engagement with primary sources, and discouraging uncredited copying and pasting (Hosseini et al., 2023; Huh, 2023).

Beyond integrity, participants highlighted the importance of authentic academic work for developing the capacity to critically assess and apply knowledge in clinical environments. Their perspectives echo long-standing views that honest, original scholarship supports the

development of professional competence and strengthens critical and analytical skills needed in practice (Ahmad et al., 2023; Papathanasiou et al., 2014). Several participants expressed concern that heavy dependence on GenAI for assignments or research could reduce the cognitive effort invested in processing and internalizing knowledge, potentially hindering the growth of analytical skills required for nursing practice when information is readily available without integrating diverse sources. In their view, such dependency could contribute to gaps in readiness for high-quality patient care by limiting opportunities to cultivate decision-making and clinical reasoning (Papathanasiou et al., 2014).

Reflecting these perceptions, participants advocated for clear institutional guidelines that define plagiarism and delineate ethical GenAI use in learning and assessment. They indicated that such standards would help ensure authentic, equitable learning experiences and safeguard academic integrity (Francis et al., 2025; Pratschke, 2024). Establishing and communicating these expectations may, in participants' view, better prepare them to become competent and ethical professionals while responsibly leveraging GenAI in their education.

Limitations

The main limitations of this study include challenges related to the generalizability of its findings to a broader population, which constrain the ability to draw wider, more comprehensive conclusions. In addition, data were collected from a single academic centre, which may limit external validity and the transferability of results to other institutional contexts. Despite these limitations, the findings remain valuable in enhancing our understanding of undergraduate nursing students' perceptions of GenAI. Another limitation is the potential for researcher bias in interpreting narrative data, as personal perspectives might have inadvertently influenced the data collection process. To mitigate this risk, the research team engaged in frequent discussions to scrutinize emergent findings and employed the technique of bracketing, which involves setting aside preconceptions to reduce bias.

The study utilized focus groups to gather data, which facilitated the collection of rich, nuanced information in the participants' own language. This method allowed participants to build on each other's responses and generate new ideas. However, the group setting may have influenced individual responses due to response bias, where participants might align their answers with perceived group norms or dominant opinions.

Additionally, the study did not specifically investigate which GenAI tool each participant used, which could have affected their experiences. Different LLM platforms may offer varied responses based on the input they receive. This lack of specificity regarding the tools used may impact the study's findings, as the diverse functionalities and interaction styles of different GenAI technologies can influence user experience significantly.

Education implications

Regarding future considerations, educational strategies should be designed to encourage students to actively explore the capabilities of GenAI and engage in reflective practices concerning its application (Pratschke, 2024). Self-reflection is an essential component that enables students to grasp both the benefits and limitations of GenAI, promoting a balanced understanding of its role in their education. By integrating GenAI into coursework and assignments, students can gain valuable hands-on experience, allowing them to better comprehend the potential educational impact of these technologies.

To facilitate this integration, specific courses covering subjects on "Anatomy and Physiology", "Pharmacology" and "Medical and Surgical in Nursing Management" could be integrated and developed within the curriculum that incorporate GenAI-related tasks. These modules would provide structured opportunities for students to interact with GenAI

tools, apply them to real-world scenarios, and critically assess their utility and limitations. Following these tasks, evaluations could be conducted to assess their effectiveness in promoting learning outcomes. These evaluations would encourage students to engage in critical thinking about the role of GenAI in enhancing educational experiences while also recognizing and understanding its constraints.

Moreover, faculty and educators should be equipped with the necessary training and resources to effectively guide students in these explorations. This support ensures that the integration of GenAI into educational practices is both meaningful and pedagogically sound. By fostering an environment where students are encouraged to critically analyze and reflect on their interactions with GenAI, educational institutions can better prepare students for future challenges in a landscape increasingly influenced by artificial intelligence.

Future research

Building on these findings, priority areas include multi-centre studies with larger, diverse samples and longitudinal cohorts to improve generalizability. Longitudinal cohort studies tracking students from entry to graduation and into clinical practice to understand how GenAI shapes self-directed learning, engagement, identification of learning needs, and professional competence over time may also add insights into this area. More rigorous trials of GenAI-integrated teaching should also assess outcomes like clinical reasoning, engagement, and integrity. Work is also needed on trust and credibility, equity and privacy, and redesigning assessments and policies to preserve originality while enabling productive GenAI use. Finally, developing validated instruments and privacy-preserving analytics will support robust, comparable evaluation across contexts.

Conclusion

In conclusion, GenAI presents substantial opportunities for enhancing students' educational experiences by offering personalized, on-demand support that enables them to explore topics more thoroughly and relate them to real-world clinical scenarios. Nonetheless, this study underscores concerns regarding academic integrity, as inappropriate use of GenAI content may lead to plagiarism and compromise academic honesty. These issues highlight important ethical considerations, underscoring the necessity for well-defined guidelines on the proper use of GenAI to preserve authenticity and originality in student work. Future educational strategies should aim to integrate GenAI into the curriculum thoughtfully, fostering self-reflection on its benefits and limitations, and creating specialized modules that promote critical thinking and ethical usage. Through these measures, students can fully leverage the educational potential of GenAI while ensuring the development of essential skills for clinical practice and professional competence.

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CRedit authorship contribution statement

Chiew-Jiat Rosalind Siah: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Junwei Vincent Lim:** Writing – review & editing, Validation, Project administration, Investigation, Formal analysis, Data curation. **Meng Meng Lyu:** Writing – review & editing, Software, Resources, Formal analysis. **Tracy Levett-Jones:** Writing – review & editing, Visualization, Validation. **Ihsan Mattar:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation.

Ethics approval statement

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